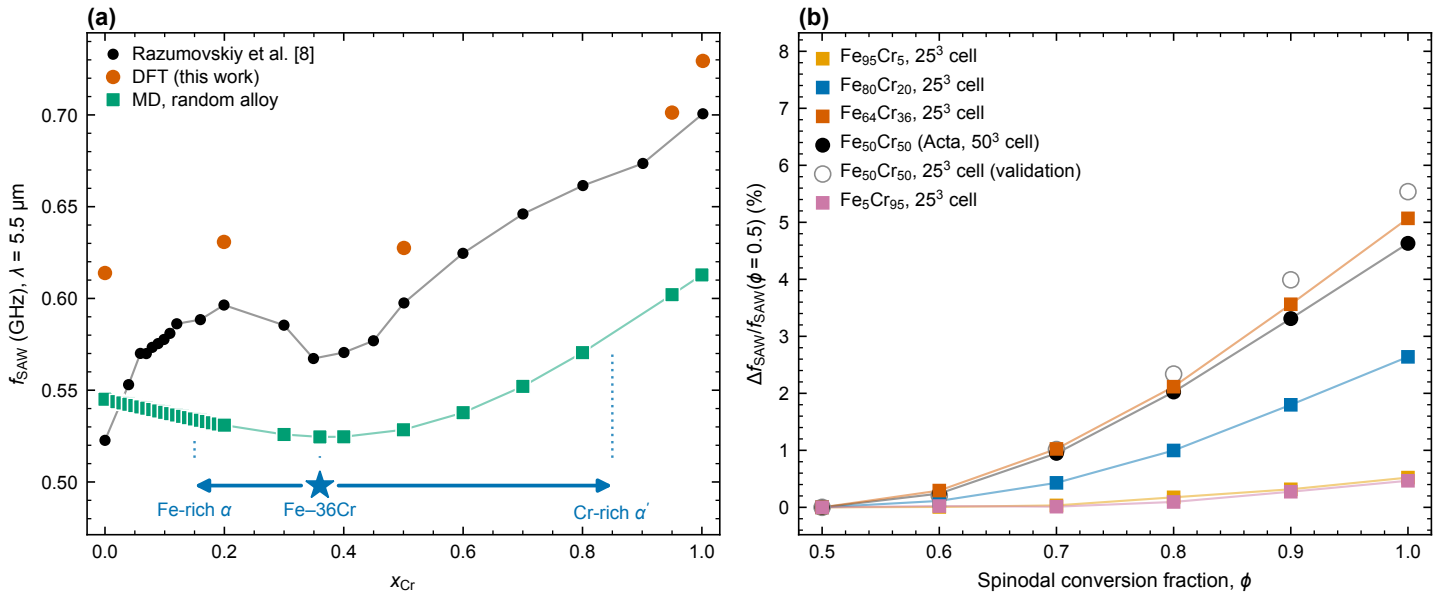
(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
$f_{SAW}$ (GHz), $\lambda$ = 5.5 µm
0.50
0.55
0.60
0.65
0.70
0.0
0.2
0.4
0.6
0.8
1.0
$x_{Cr}$
Fe-rich $\alpha$
Fe–36Cr
Cr-rich $\alpha'$
(b)
$Fe_{95}Cr_5$, $25^3$ cell
$Fe_{80}Cr_{20}$, $25^3$ cell
$Fe_{64}Cr_{36}$, $25^3$ cell
$Fe_{50}Cr_{50}$ (Acta, $50^3$ cell)
$Fe_{50}Cr_{50}$, $25^3$ cell (validation)
$Fe_5Cr_{95}$, $25^3$ cell
$\Delta f_{SAW}/f_{SAW}(\phi = 0.5)$ (%)
0
1
2
3
4
5
6
7
8
0.5
0.6
0.7
0.8
0.9
1.0
Spinodal conversion fraction, $\phi$